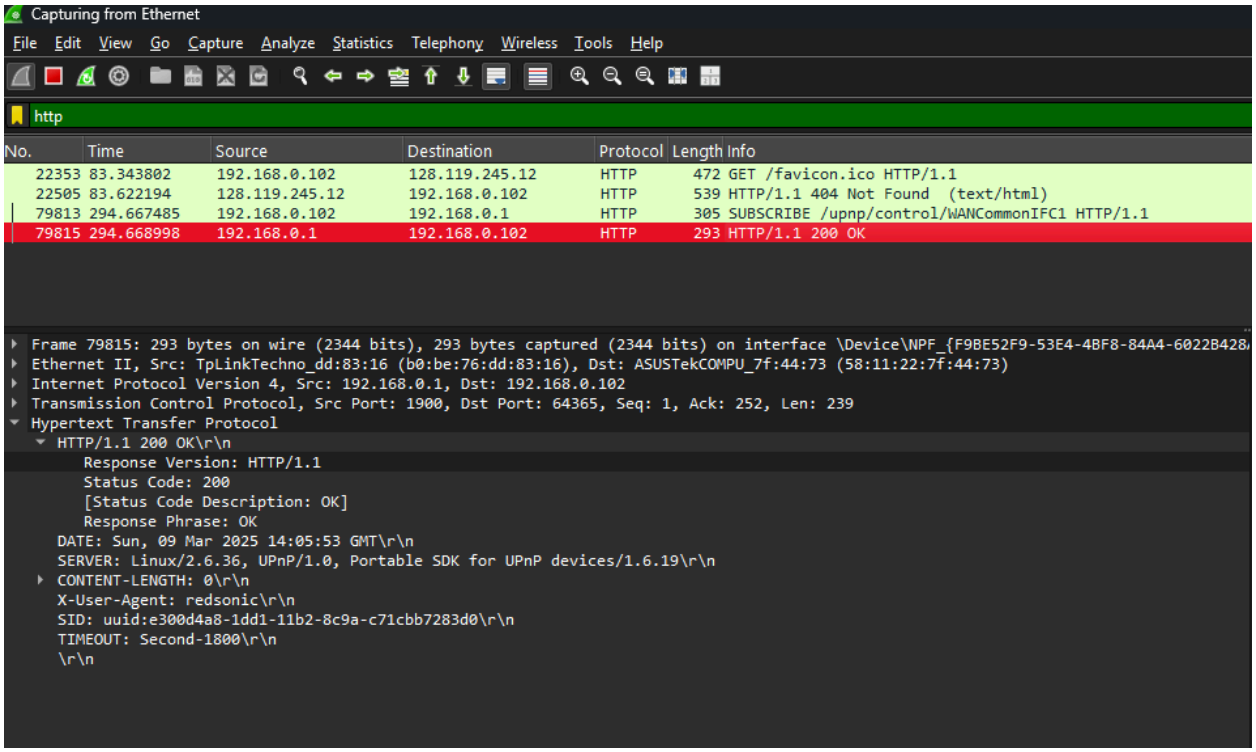
1. จุติณัฏฐ์ รัตนะมงคลกุล 6609650228
2. รพินทร์ นะราช 6609650624

Wireshark Lab 01 -- HTTP
Assignment

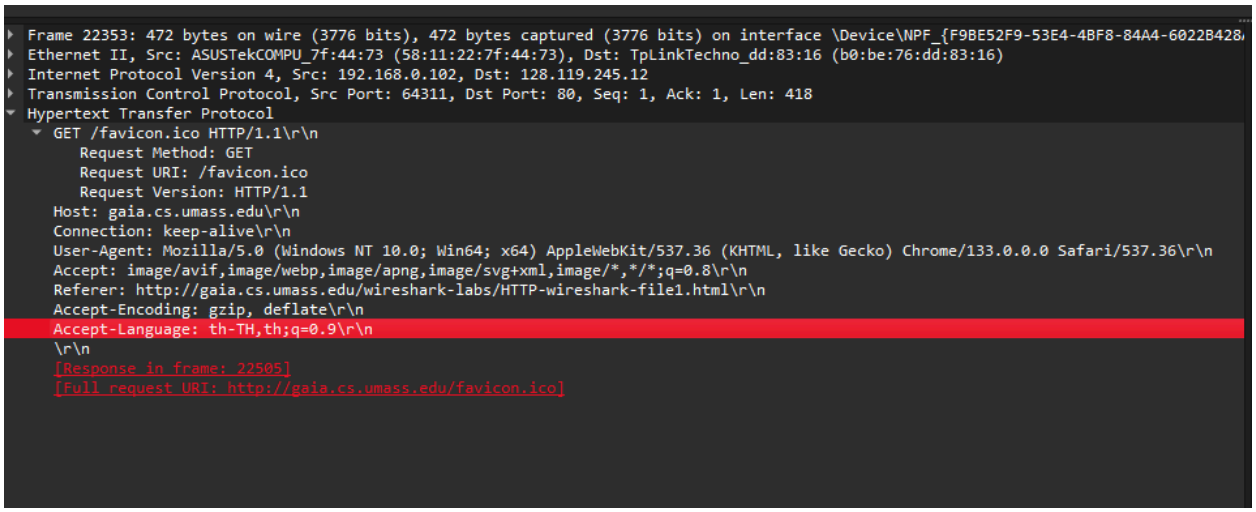
1. The Basic HTTP GET/response interaction

1. Is your browser running HTTP version 1.0, 1.1, or 2? What version of HTTP is the server running?



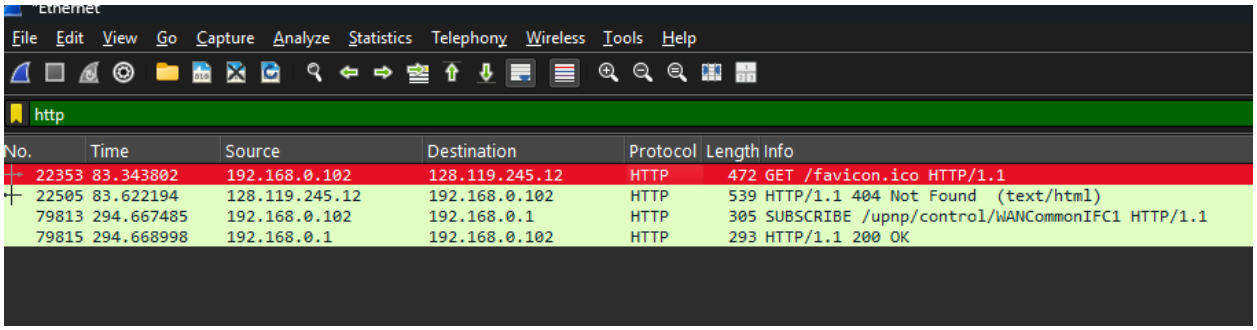
ตอบ: บราวเซอร์รัน HTTP บนเวอร์ชั่น 1.1

2. What languages (if any) does your browser indicate that it can accept to the server?



ตอบ: th-TH, th;q=0.9\r\n เป็นภาษาไทย และ Quality Factor ที่ 0.9

3. What is the IP address of your computer? What is the IP address of the gaia.cs.umass.edu server?

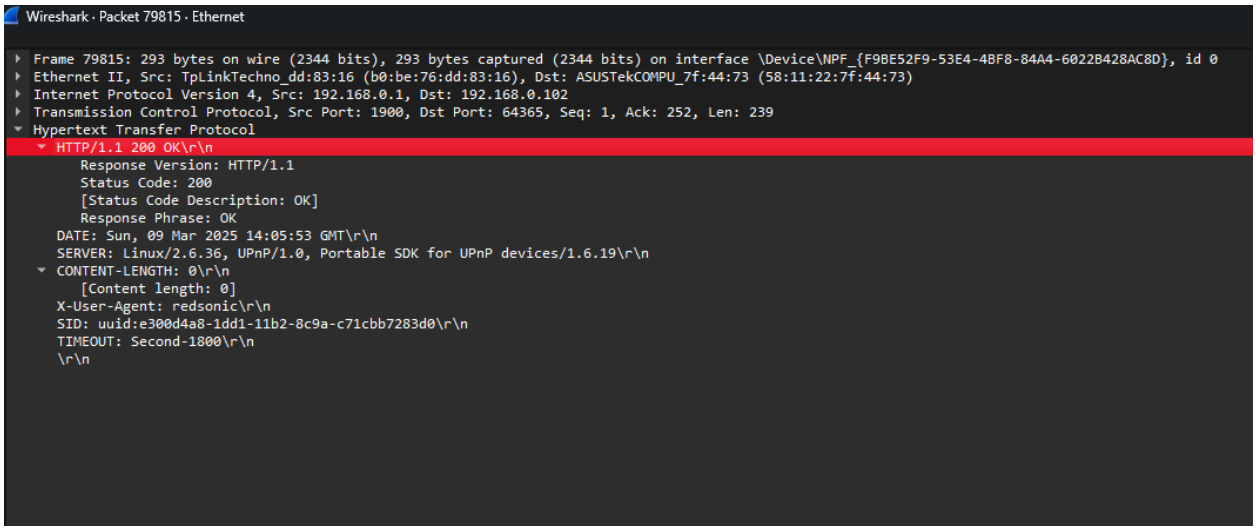


ตอบ:

IP ของคอมพิวเตอร์คือ 192.168.0.102 และ

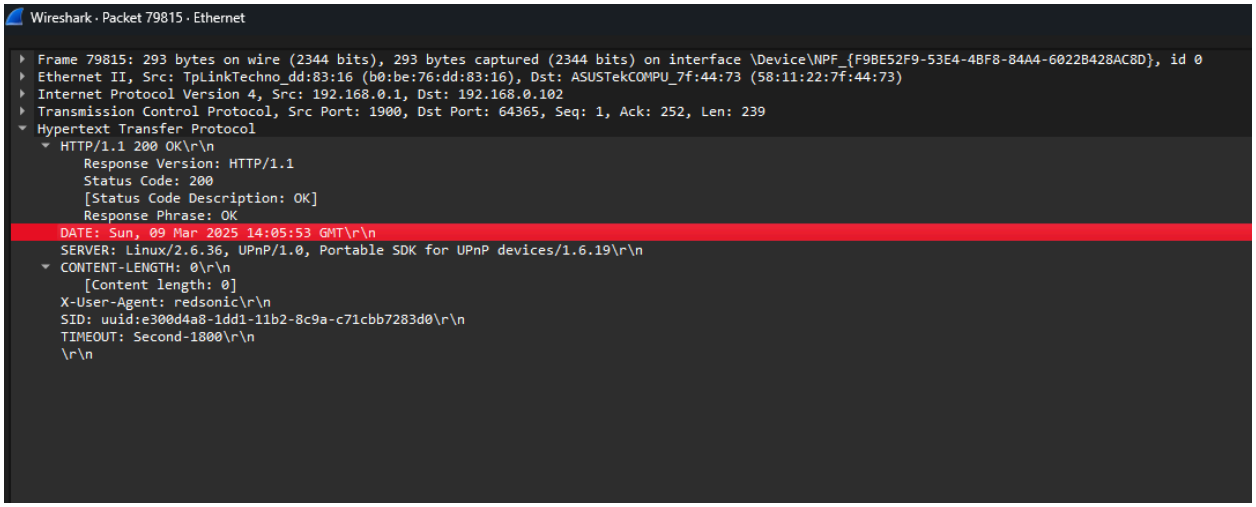
IP ของเซิร์ฟเวอร์ gaia.cs.umass.edu คือ 128.119.245.12

4. What is the status code returned from the server to your browser?



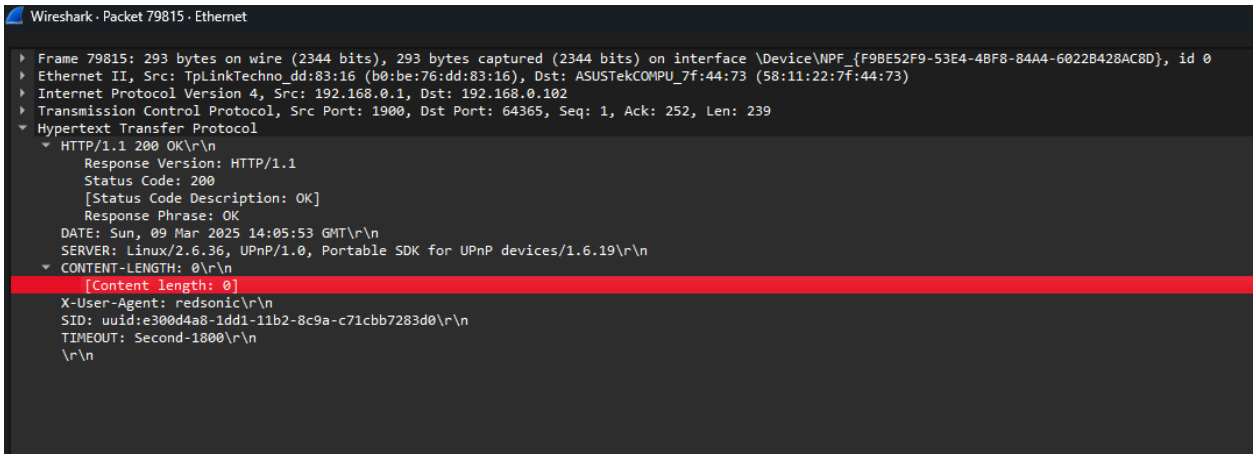
ตอบ: 200 OK

5. When was the HTML file that you are retrieving last modified at the server?



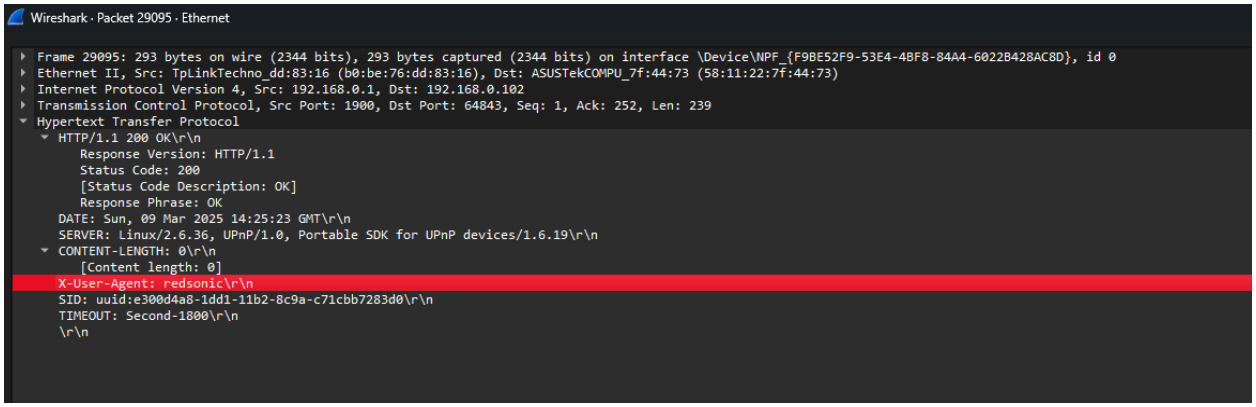
ตอบ: Sun, 09 Mar 2025 14:05:53 GMT

6. How many bytes of content are being returned to your browser?



ตอบ: 0 byte ส่งกลับมาที่บราวเซอร์

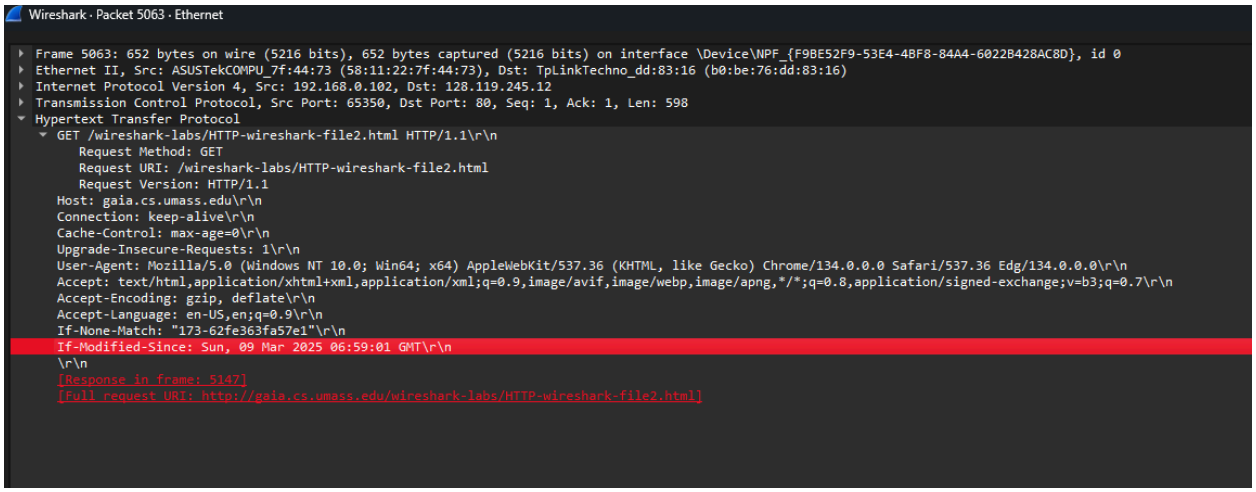
7. By inspecting the raw data in the packet content window, do you see any headers within the data that are not displayed in the packet-listing window? If so, name one.



ตอบ: พบเฮดเดอร์ X-User-Agent ในหน้าต่าง packet content ซึ่งไม่ได้แสดงอยู่ใน packet-listing window

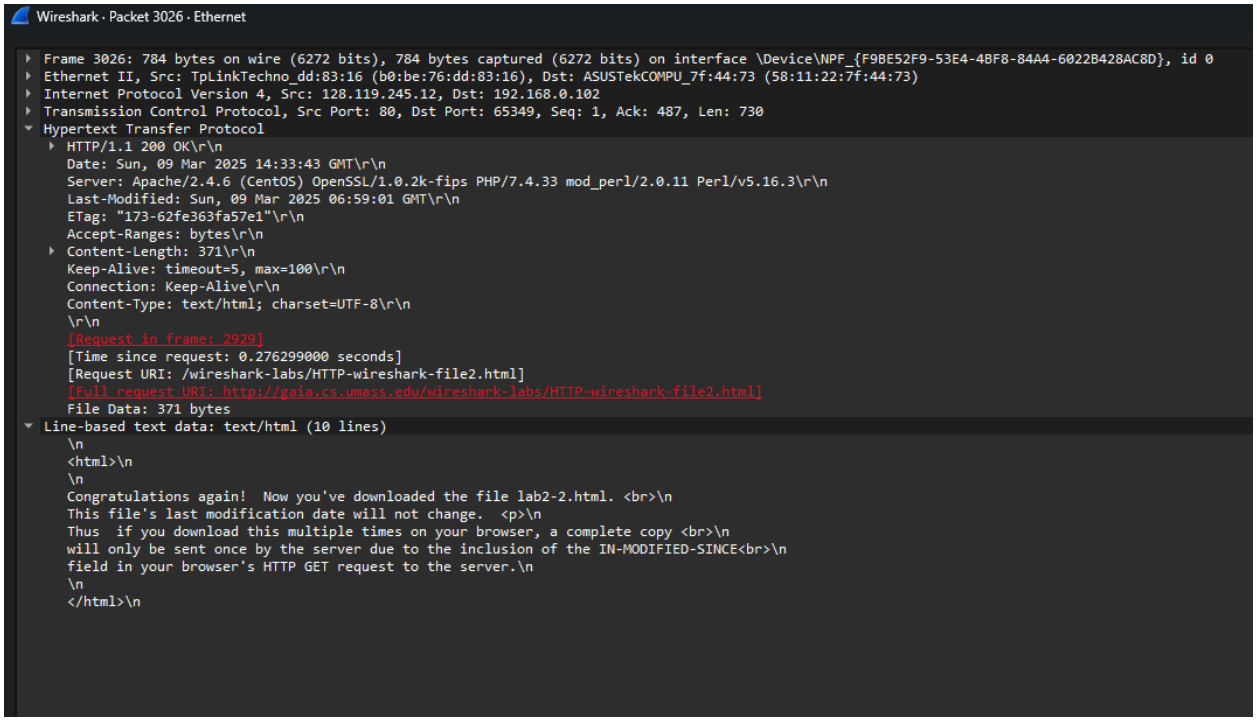
2. The HTTP CONDITIONAL GET/response interaction

8. Inspect the contents of the first HTTP GET request from your browser to the server. Do you see an “IF-MODIFIED-SINCE” line in the HTTP GET?



ตอบ: Modified since Sun, 09 Mar 2025 06:59:01

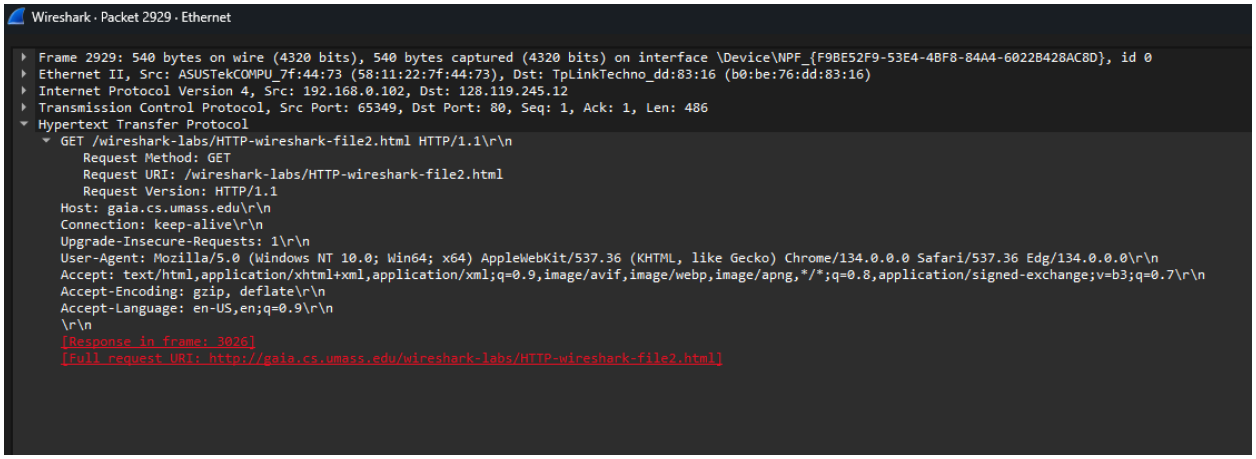
9. Inspect the contents of the server response. Did the server explicitly return the contents of the file? How can you tell?



ตอบ: มีเนื้อหาจากไฟล์ HTML
\n
<html>\n
\n
Congratulations again! Now you've downloaded the file lab2-2.html.
\n
This file's last modification date will not change. <p>\n
Thus if you download this multiple times on your browser, a complete copy

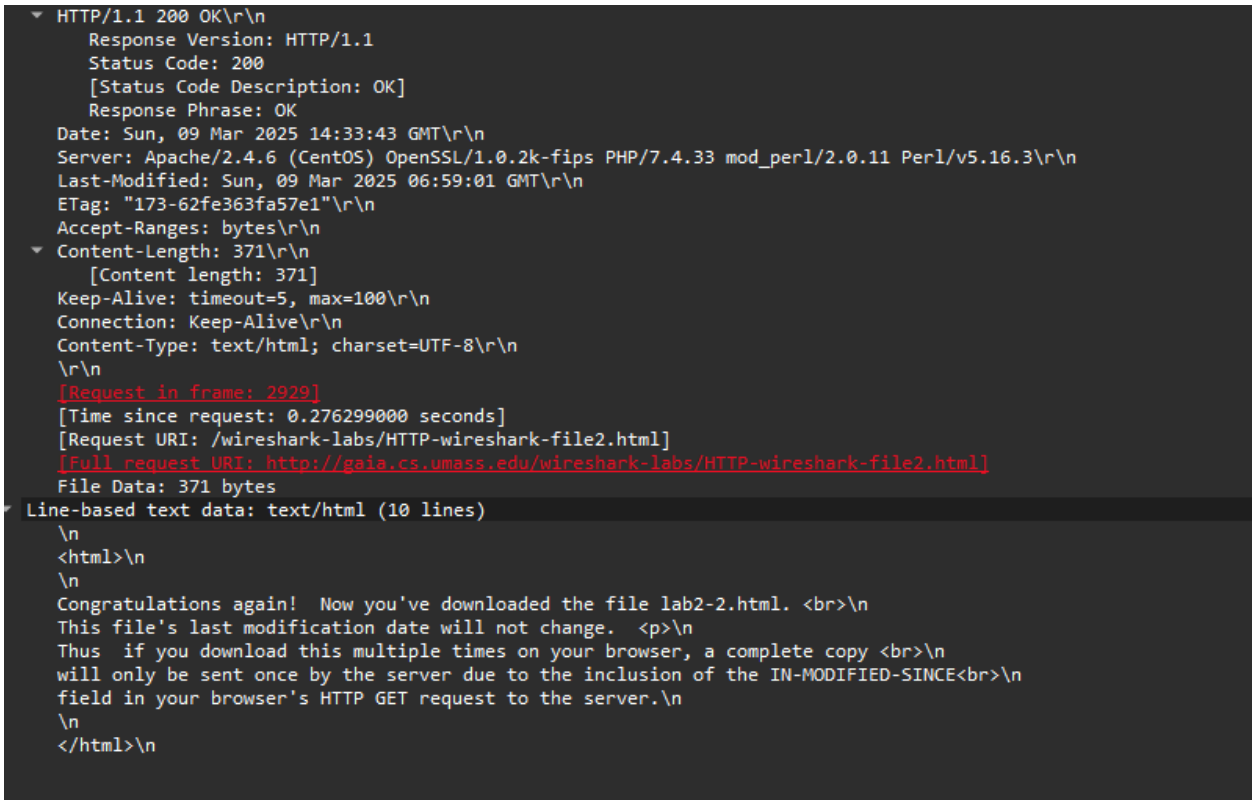
\n
will only be sent once by the server due to the inclusion of the
IN-MODIFIED-SINCE
\n
field in your browser's HTTP GET request to the server.\n
\n
</html>\n

10. Now inspect the contents of the second HTTP GET request from your browser to the server. Do you see an “IF-MODIFIED-SINCE:” line in the HTTP GET6 ? If so, what information follows the “IF-MODIFIED-SINCE:” header?



ตอบ: ไม่มี IF-MODIFIED-SINCE: ใน HTTP GET อันที่ 2

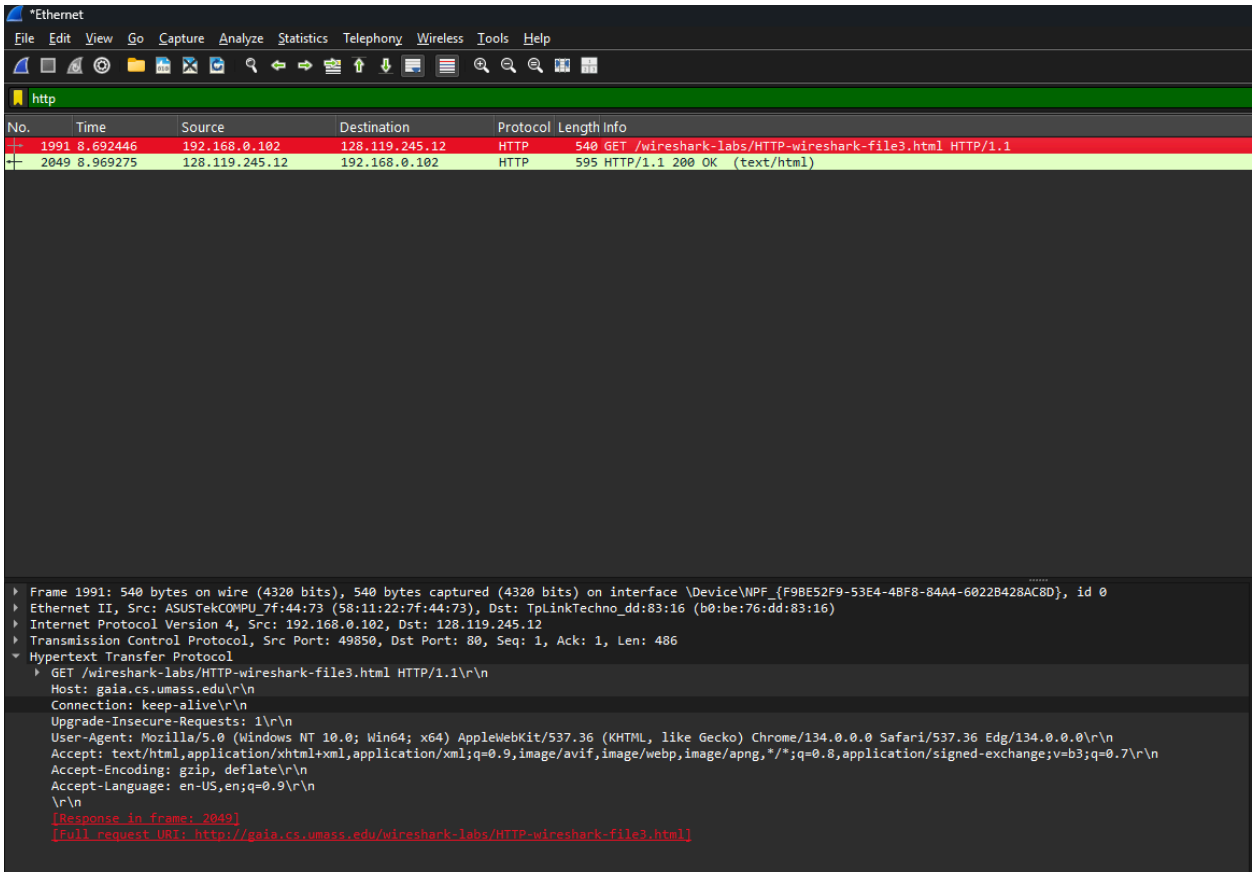
11. What is the HTTP status code and phrase returned from the server in response to this second HTTP GET? Did the server explicitly return the contents of the file? Explain.



ตอบ: เซิร์ฟเวอร์ส่ง HTTP GET กลับมาครั้งที่สองว่า ไฟล์ที่ร้องขอถูกส่งกลับมาทั้งหมด โดยไม่มีการใช้แคช

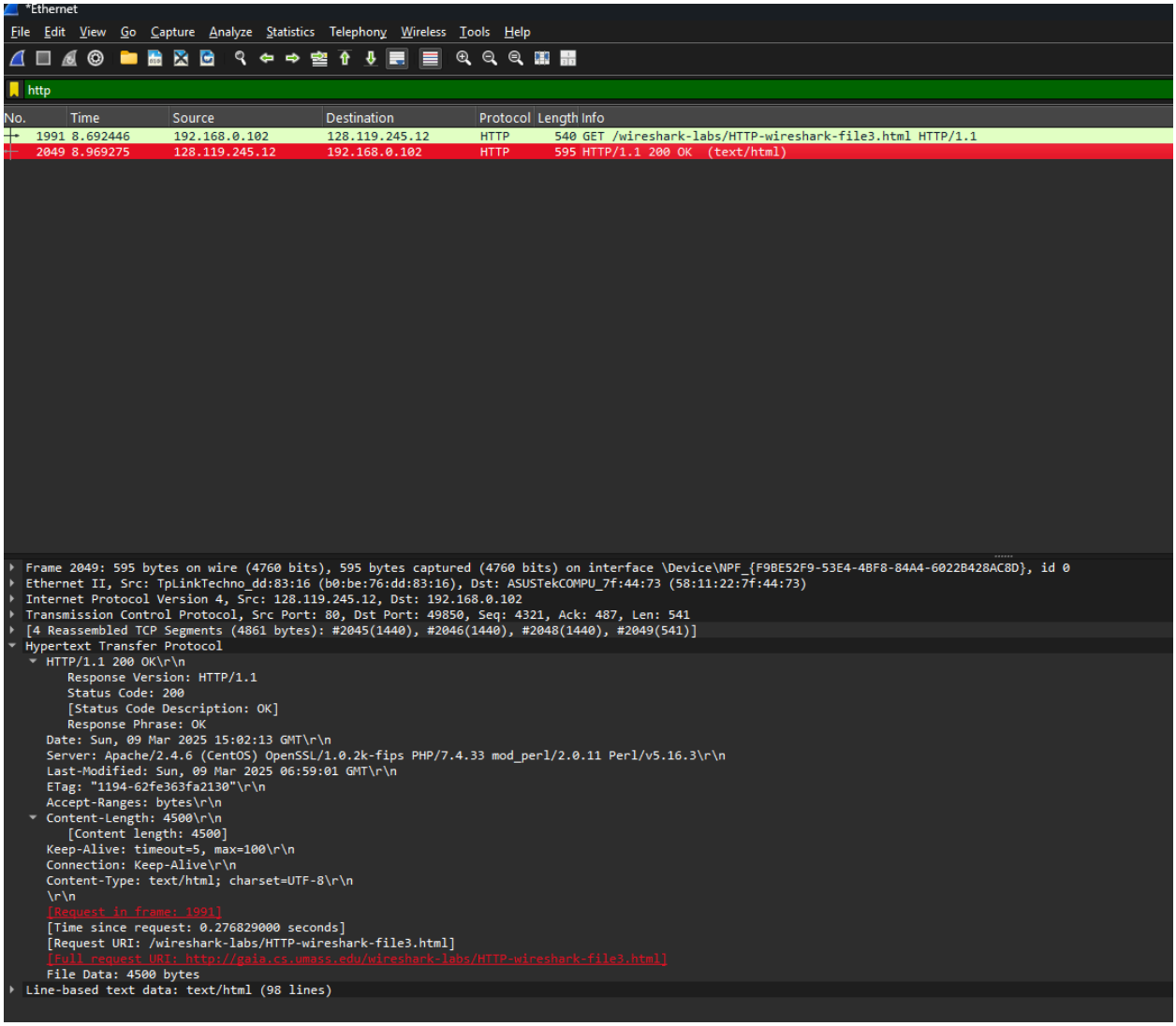
3. Retrieving Long Documents

12. How many HTTP GET request messages did your browser send? Which packet number in the trace contains the GET message for the Bill or Rights?



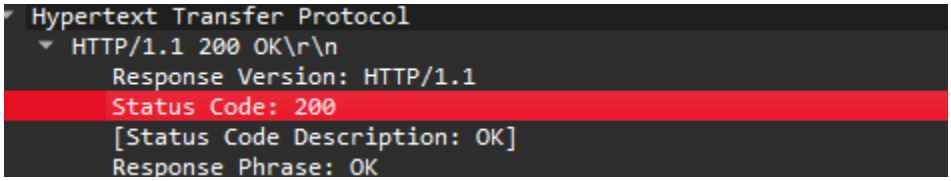
ตอบ: เบราว์เซอร์ส่ง 1 HTTP GET Request ไปที่เซิร์ฟเวอร์

13. Which packet number in the trace contains the status code and phrase associated with the response to the HTTP GET request?



ตอบ: หมายเลขแพ็กเก็ตของ HTTP Response 2049

14. What is the status code and phrase in the response?



ตอบ: Status Code เป็น 200 OK

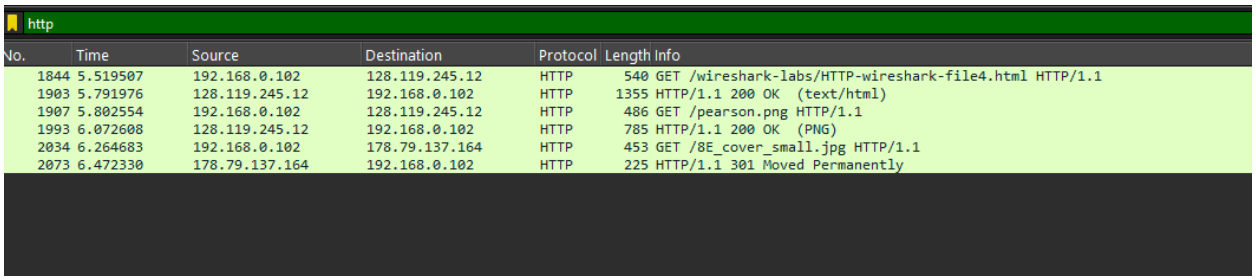
15. How many data-containing TCP segments were needed to carry the single HTTP response and the text of the Bill of Rights?

```
▾ Transmission Control Protocol, Src Port: 80, Dst Port: 49850, Seq: 4321, Ack: 487, Len: 541
    Source Port: 80
    Destination Port: 49850
    [Stream index: 23]
    [Stream Packet Number: 10]
  ▸ [Conversation completeness: Complete, WITH_DATA (31)]
    [TCP Segment Len: 541]
    Sequence Number: 4321    (relative sequence number)
    Sequence Number (raw): 693752829
    [Next Sequence Number: 4862    (relative sequence number)]
    Acknowledgment Number: 487    (relative ack number)
    Acknowledgment number (raw): 2557274321
    0101 .... = Header Length: 20 bytes (5)
  ▸ Flags: 0x018 (PSH, ACK)
    Window: 237
    [Calculated window size: 30336]
    [Window size scaling factor: 128]
    Checksum: 0x2961 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
  ▸ [Timestamps]
  ▸ [SEQ/ACK analysis]
    TCP payload (541 bytes)
    TCP segment data (541 bytes)
```

ตอบ: Transmission Control Protocol, Src Port: 80, Dst Port: 49850, Seq: 4321, Ack: 487, Len: 541

4. HTML Documents with Embedded Objects

16. How many HTTP GET request messages did your browser send? To which Internet addresses were these GET requests sent?



http

No.	Time	Source	Destination	Protocol	Length	Info
1844	5.519507	192.168.0.102	128.119.245.12	HTTP	540	GET /wireshark-labs/HTTP-wireshark-file4.html HTTP/1.1
1903	5.791976	128.119.245.12	192.168.0.102	HTTP	1355	HTTP/1.1 200 OK (text/html)
1907	5.802554	192.168.0.102	128.119.245.12	HTTP	486	GET /pearson.png HTTP/1.1
1993	6.072608	128.119.245.12	192.168.0.102	HTTP	785	HTTP/1.1 200 OK (PNG)
2034	6.264683	192.168.0.102	178.79.137.164	HTTP	453	GET /8E_cover_small.jpg HTTP/1.1
2073	6.472330	178.79.137.164	192.168.0.102	HTTP	225	HTTP/1.1 301 Moved Permanently

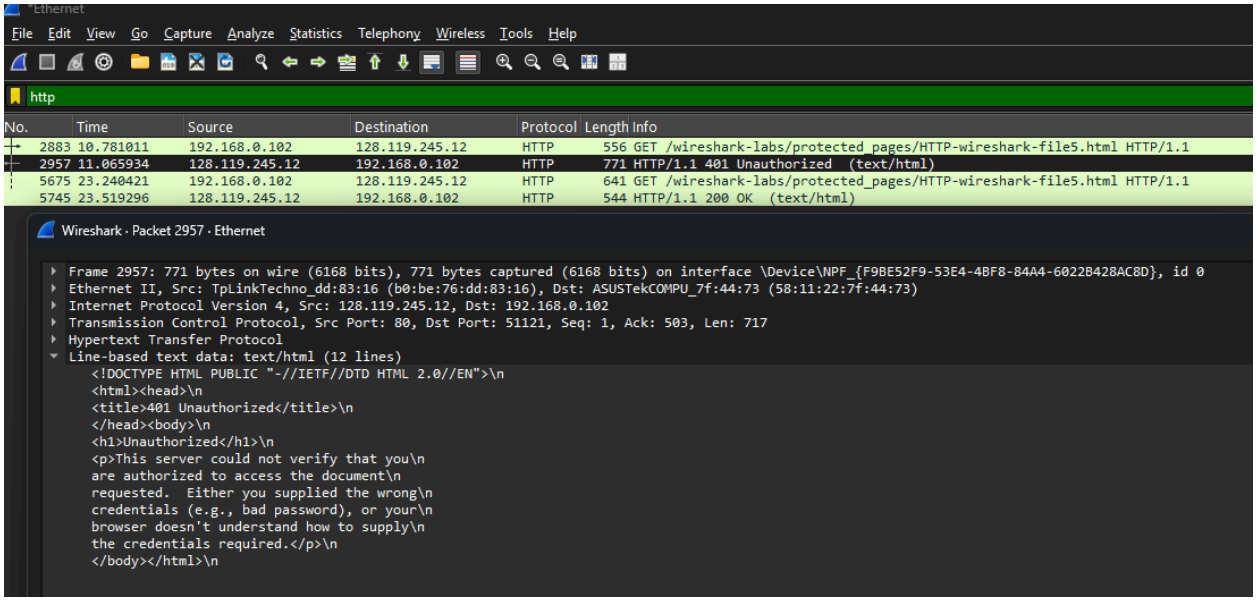
ตอบ: มี 3 HTTP GET Request (192.168.0.102)

17. Can you tell whether your browser downloaded the two images serially, or whether they were downloaded from the two websites in parallel? Explain.

ตอบ: มีการขอภาพจากแต่ละเว็บไซต์เกือบพร้อมกัน แต่ไม่ใช่ในเวลาเดียวกันเป๊ะ ๆ มีการขอไฟล์ pearson.png จากเซิร์ฟเวอร์ 128.119.245.12 แล้วอีกไฟล์ 8E_cover_small.jpg จากเซิร์ฟเวอร์ 178.79.137.164 ในช่วงเวลาห่างกันเล็กน้อย

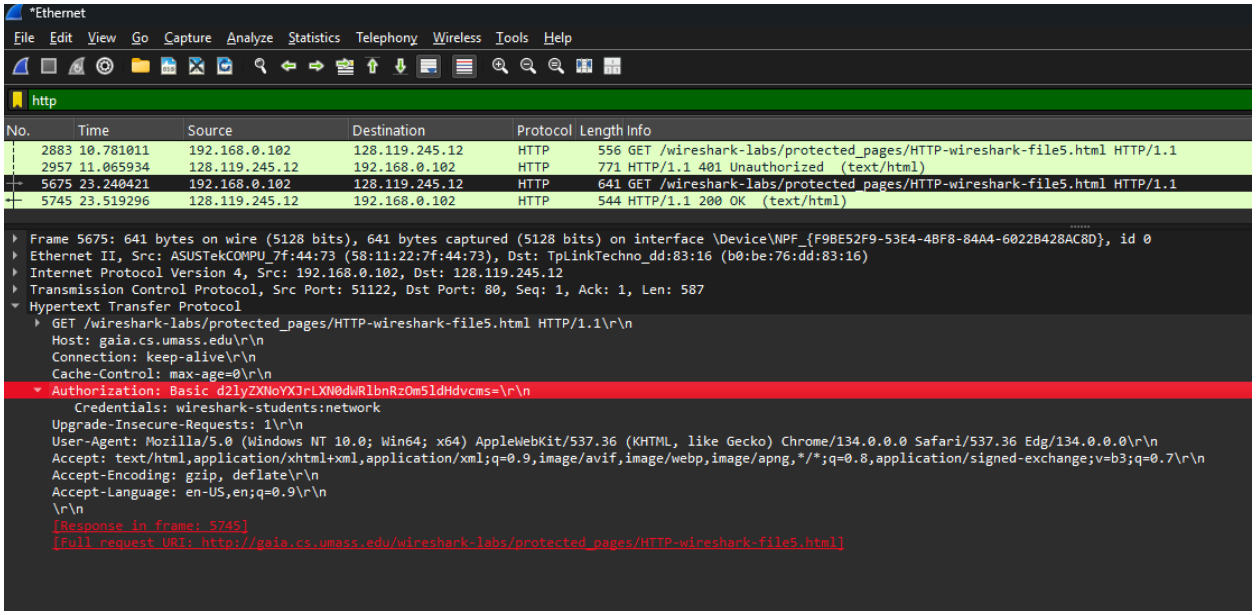
5 HTTP Authentication

18. What is the server's response (status code and phrase) in response to the initial HTTP GET message from your browser?



ตอบ: แพ็กเก็ตที่ No. 2957 ซึ่งเป็น HTTP Response ที่เซิร์ฟเวอร์ส่งกลับมาหลังจากที่เบราว์เซอร์ส่ง HTTP GET status code: HTTP/1.1 401 Unauthorized

19. When your browser's sends the HTTP GET message for the second time, what new field is included in the HTTP GET message?



ตอบ: Authorization: Basic d2lyZXNoYXJrLXN0dWRlbnRzOm5ldHdvcms=

ดูจากแพ็กเก็ตที่ No. 5675 ซึ่งเป็น HTTP GET ที่ส่งไปอีกครั้งหลังจากได้รับ 401 Unauthorized หลังจากนั้น เซิร์ฟเวอร์ตอบกลับมาด้วย HTTP/1.1 200 OK ในแพ็กเก็ตที่ No. 5745 หลังจากยืนยันตัวตนสำเร็จ